

# DNS (Domain Name System) – Clear Notes

## What is DNS?

- DNS stands for **Domain Name System**.
- It acts like the **phonebook of the internet**.
- Humans remember domain names easily (example: `tryhackme.com`).
- Computers communicate using **IP addresses** (example: `104.26.10.229`).
- DNS converts domain names into IP addresses.

### Example

tryhackme.com → 104.26.10.229

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## Structure of a Domain Name

### 1. TLD (Top-Level Domain)

The **right-most** part of a domain name.

#### Example

tryhackme.com

↑  
TLD

### Types of TLDs

#### a) gTLD (Generic Top-Level Domain)

Used based on purpose.

Examples:

- `.com` → Commercial
- `.org` → Organization
- `.edu` → Education
- `.gov` → Government

Modern gTLDs:

- `.online`
  - `.club`
  - `.website`
  - `.biz`
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## b) ccTLD (Country Code Top-Level Domain)

Used for countries/regions.

Examples:

- `.in` → India
  - `.uk` → United Kingdom
  - `.ca` → Canada
  - `.co.uk` → UK commercial sites
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# 2. Second-Level Domain (SLD)

The part before the TLD.

## Example

tryhackme.com

^^^^^^^^^^

Second-Level Domain

## Rules

- Maximum: **63 characters**
- Allowed:
  - `a-z`
  - `0-9`
  - `-` (hyphen)

## Restrictions

- Cannot start/end with -
  - No consecutive hyphens (--)
- 

## 3. Subdomain

A subdomain appears on the **left side** of the SLD.

### Example

admin.tryhackme.com  
^^^^

Subdomain

### More Examples

mail.google.com  
api.github.com  
jupiter.servers.tryhackme.com

### Rules

- Same rules as SLD
- Max length:
  - Each label: 63 characters
  - Full domain: 253 characters

### Important

- Unlimited subdomains can be created.
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## Common DNS Record Types

### 1. A Record

Maps a domain to an **IPv4 address**.

### Example

example.com → 104.26.10.229

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## 2. AAAA Record

Maps a domain to an **IPv6 address**.

### Example

example.com → 2606:4700:20::681a:be5

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## 3. CNAME Record (Canonical Name)

Points one domain to another domain.

### Example

store.tryhackme.com → shops.shopify.com

### Process

1. DNS resolves `store.tryhackme.com`
  2. It returns `shops.shopify.com`
  3. Another DNS lookup happens for Shopify's IP
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## 4. MX Record (Mail Exchange)

Specifies mail servers for a domain.

### Example

tryhackme.com → alt1.aspmx.l.google.com

### Features

- Used for email delivery
- Includes **priority values**
- Lower number = higher priority

### Purpose

If the main mail server fails, backup servers handle emails.

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## 5. TXT Record

Stores text information.

### Common Uses

- SPF records
- DMARC policies
- Domain verification
- Email security

### Examples

\_acme-challenge.example.com TXT "token\_value\_here"

@ TXT "v=spf1 ip4:192.0.2.0/24 include:\_spf.google.com ~all"

\_dmarc.example.com TXT "v=DMARC1; p=reject"

@ TXT "MS=ms12345678"

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# DNS Resolution Process (How DNS Works)

When you visit a website, DNS follows several steps.

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## Step 1: Check Local Cache

Your computer first checks:

- Browser cache
- OS cache

If IP already exists:

Domain found → Use saved IP

No further request needed.

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## Step 2: Query Recursive DNS Server

If not found locally:

- Request goes to a **Recursive DNS Resolver**
- Usually provided by:
  - ISP
  - Google DNS
  - Cloudflare DNS

### Examples

8.8.8.8 → Google DNS

1.1.1.1 → Cloudflare DNS

The resolver also checks its cache.

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## Step 3: Root DNS Server

If cache miss occurs:

- Resolver asks a **Root DNS Server**

### Root Server Role

It directs requests to the correct TLD server.

### Example

For:

www.tryhackme.com

Root server says:

Ask the .com TLD server

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## Step 4: TLD Server

The `.com` TLD server knows:

- Which authoritative server stores records for the domain.

### Response Example

tryhackme.com nameserver:  
kip.ns.cloudflare.com  
uma.ns.cloudflare.com

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## Step 5: Authoritative DNS Server

This server stores actual DNS records.

It returns:

- A record
- MX record
- TXT record
- etc.

### Example

tryhackme.com → 104.26.10.229

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## Step 6: Response Returned

The result goes back:

Authoritative Server

↓

Recursive Resolver

↓

Your Computer

Then your browser connects to the website.

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# DNS Caching

DNS responses are temporarily stored to improve speed.

## TTL (Time To Live)

TTL tells how long DNS data should stay cached.

### Example

TTL = 3600

Means:

Cache for 1 hour

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# Why DNS Caching is Important

## Advantages

- Faster browsing
  - Reduced DNS traffic
  - Less load on DNS servers
  - Better performance
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## Simple DNS Flow Summary

User enters domain



Check Local Cache



Recursive DNS Resolver



Root DNS Server



TLD Server



Authoritative DNS Server  
↓  
IP Address Returned  
↓  
Browser connects to Website

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## Quick Revision Table

| Component            | Purpose                               |
|----------------------|---------------------------------------|
| DNS                  | Converts domain names to IP addresses |
| TLD                  | Right-most part (.com, .org)          |
| SLD                  | Main domain name                      |
| Subdomain            | Additional section before domain      |
| A Record             | Maps to IPv4                          |
| AAAA Record          | Maps to IPv6                          |
| CNAME                | Alias to another domain               |
| MX Record            | Mail server information               |
| TXT Record           | Stores text/security info             |
| Recursive Resolver   | Finds DNS answers                     |
| Root Server          | Directs to TLD server                 |
| TLD Server           | Directs to authoritative server       |
| Authoritative Server | Stores actual DNS records             |
| TTL                  | Cache duration                        |

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## Important Interview Points

### Frequently Asked Questions

## **What is DNS?**

DNS converts human-readable domain names into IP addresses.

## **Difference between A and AAAA records?**

- A → IPv4
- AAAA → IPv6

## **What is a CNAME record?**

An alias pointing one domain to another domain.

## **What is an MX record used for?**

Handling email routing.

## **What is TTL in DNS?**

Time duration for caching DNS records.

## **What is an authoritative DNS server?**

The server storing actual DNS records for a domain.

## **What is the role of the root server?**

Redirects queries to the correct TLD server.